

B6 Condensed-Matter Physics — Model Solutions, Trinity 2021

Question 1 — Powder neutron diffraction of cubic RbMnF₃

Problem. Cubic RbMnF₃ powder, neutrons of $E = 25$ meV. Rings at $2\theta = 24.6^\circ, 35.1^\circ, 43.4^\circ, 50.5^\circ$. Find d_{hkl} , Miller indices, lattice type and accurate a (≈ 0.4 nm). Below $T_c = 83$ K two new rings; one at 21.3° . Mechanism, predict the second.

(a) Why neutrons rather than X-rays

Neutrons carry a magnetic moment that couples to unpaired electron spin density.

Use neutrons to detect magnetic

(b) Lattice-plane spacings

de Broglie wavelength from $E = p^2/2m_n$:

$$\lambda = \frac{h}{\sqrt{2m_n E}}.$$

Convert energy:

$$E = 25 \times 10^{-3} \times 1.602 \times 10^{-19} \text{ J} = 4.005 \times 10^{-21} \text{ J}.$$

Compute momentum:

$$\sqrt{2m_n E} = \sqrt{2 \cdot 1.67493 \times 10^{-27} \cdot 4.005 \times 10^{-21}} = 3.663 \times 10^{-24} \text{ kg m s}^{-1}.$$

Wavelength:

$$\lambda = \frac{6.626 \times 10^{-34}}{3.663 \times 10^{-24}} = 0.1809 \text{ nm}.$$

Bragg's law:

$$d_{hkl} = \frac{\lambda}{2 \sin \theta}, \quad \theta = \frac{1}{2}(2\theta). \quad (1)$$

Tabulate:

2θ	θ	$\sin \theta$	d_{hkl}
24.6°	12.30°	0.2130	0.4247 nm
35.1°	17.55°	0.3015	0.3000 nm
43.4°	21.70°	0.3697	0.2447 nm
50.5°	25.25°	0.4266	0.2120 nm

$$d_1=0.425 \text{ nm}, d_2=0.300 \text{ nm}, d_3=0.245 \text{ nm}, d_4=0.212 \text{ nm}.$$

(2)

(c) Miller indices, lattice type, and accurate a

Cubic spacing:

$$d_{hkl} = \frac{a}{\sqrt{N}}, \quad N \equiv h^2 + k^2 + l^2. \quad (3)$$

Compute $N = (a/d)^2$ with $a \approx 0.4$ nm:

$$N \approx 0.886, 1.778, 2.671, 3.560.$$

Identify $N = 1, 2, 3, 4$ after small rescaling of a .

Selection rules: $N = 1$ allowed rules out FCC (needs all-even or all-odd); $N = 2$ allowed rules out BCC (needs $h + k + l$ even, but only the larger $N = 2, 4, 6 \dots$ pattern). Hence simple cubic.

Miller indices in order: (100), (110), (111), (200).

Refit $a = d\sqrt{N}$:

$$a_1 = 0.4247 \cdot 1 = 0.4247 \text{ nm.}$$

$$a_2 = 0.3000 \cdot \sqrt{2} = 0.4243 \text{ nm.}$$

$$a_3 = 0.2447 \cdot \sqrt{3} = 0.4239 \text{ nm.}$$

$$a_4 = 0.2120 \cdot 2 = 0.4240 \text{ nm.}$$

Average:

$$\boxed{a = 0.4242(4) \text{ nm.}} \quad (4)$$

(d) Field changes ring intensity but not angle

Bragg angles depend only on d_{hkl} ; field-induced lattice distortion (magnetostriction) is negligible. Neutron structure factor splits into nuclear and magnetic parts:

$$F_{hkl} = F_{hkl}^{\text{nuc}} + F_{hkl}^{\text{mag}}(\hat{\mathbf{B}}).$$

Field polarises the Mn^{2+} paramagnetic moments, modifying $\mathbf{M}_{\perp}(\mathbf{Q})$ and hence $|F_{hkl}|^2$. Field aligns Mn^{2+} moments

(e) New rings below T_c : G-type antiferromagnetism

Mechanism: nearest-neighbour Mn moments alternate in sign on the simple-cubic Mn sublattice (G-type AFM). Magnetic unit cell is $2a \times 2a \times 2a$.

Magnetic reflections at $\mathbf{Q} = (2\pi/a)(h', k', l')$ with h', k', l' half-odd-integers; lowest-order $N_{\text{mag}} = h'^2 + k'^2 + l'^2$.

Smallest N_{mag} : $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ gives $N_{\text{mag}} = 3/4$. Spacing:

$$d^* = \frac{a}{\sqrt{3/4}} = 0.4898 \text{ nm.}$$

Bragg angle:

$$\sin \theta = \frac{\lambda}{2d^*} = \frac{0.1809}{2 \cdot 0.4898} = 0.1847.$$

$$2\theta = 21.3^\circ,$$

matching the observation. G-type AFM ordering; magnetic period $2a$.

Next allowed magnetic reflection: $(\frac{3}{2}, \frac{1}{2}, \frac{1}{2})$, $N_{\text{mag}} = 11/4$. Spacing:

$$d^{**} = \frac{a}{\sqrt{11/4}} = \frac{0.4242 \cdot 2}{\sqrt{11}} \text{ nm} = 0.2558 \text{ nm}.$$

Bragg angle:

$$\sin \theta = \frac{0.1809}{2 \cdot 0.2558} = 0.3537.$$

$$\boxed{2\theta_{2\text{nd}} \approx 41.4^\circ.} \quad (5)$$

Slots between (110) at 35.1° and (111) at 43.4° , consistent with the observation window.

Question 2 — Crystal structure and phonons of AlP / Si

Problem. AlP plan view (zinc-blende). (a) Bravais lattice, lattice points, basis. (b) Reciprocal-lattice formula and derive the AlP reciprocal lattice. (c) Acoustic/optical phonon counts; longitudinal/transverse along [100] and number of distinct acoustic velocities. (d) For Si ($a = 0.5431 \text{ nm}$, $v_L = 8433 \text{ m s}^{-1}$ along [100]) estimate $\hbar\omega_{\text{LA}}$ at $\mathbf{k} = (2\pi/a, 0, 0)$. (e) Phonon-spectrum difference between AlP and Si at this $\mathbf{k}$.

(a) Bravais lattice and basis

Al sites at corners and all six face centres: FCC. P sites at $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})a$ and the three FCC translates: same FCC lattice shifted by $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})a$. This is the zinc-blende structure.

Conventional-cell FCC lattice points:

$$(0, 0, 0), (\frac{1}{2}, \frac{1}{2}, 0), (\frac{1}{2}, 0, \frac{1}{2}), (0, \frac{1}{2}, \frac{1}{2}).$$

$$\boxed{\text{Bravais: FCC. Basis: Al at } (0, 0, 0), \text{ P at } (\frac{1}{4}, \frac{1}{4}, \frac{1}{4})a.} \quad (6)$$

(b) Reciprocal lattice

General formula ($\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$):

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}, \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}. \quad (7)$$

FCC primitive vectors:

$$\mathbf{a}_1 = \frac{a}{2}(\hat{\mathbf{y}} + \hat{\mathbf{z}}), \quad \mathbf{a}_2 = \frac{a}{2}(\hat{\mathbf{z}} + \hat{\mathbf{x}}), \quad \mathbf{a}_3 = \frac{a}{2}(\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

Compute the cross product:

$$\mathbf{a}_2 \times \mathbf{a}_3 = \frac{a^2}{4}(\hat{\mathbf{z}} + \hat{\mathbf{x}}) \times (\hat{\mathbf{x}} + \hat{\mathbf{y}}) = \frac{a^2}{4}(\hat{\mathbf{y}} - \hat{\mathbf{x}} + \hat{\mathbf{z}}).$$

Primitive-cell volume:

$$\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3) = \frac{a^3}{8}(\hat{\mathbf{y}} + \hat{\mathbf{z}}) \cdot (\hat{\mathbf{y}} - \hat{\mathbf{x}} + \hat{\mathbf{z}}) = \frac{a^3}{4}.$$

Substitute into (7):

$$\mathbf{b}_1 = \frac{2\pi}{a}(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}), \quad \mathbf{b}_2 = \frac{2\pi}{a}(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}), \quad \mathbf{b}_3 = \frac{2\pi}{a}(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}).$$

Recognise BCC primitives:

$$\mathbf{b}_1 + \mathbf{b}_2 = \frac{4\pi}{a}\hat{\mathbf{z}}, \quad \mathbf{b}_2 + \mathbf{b}_3 = \frac{4\pi}{a}\hat{\mathbf{x}}, \quad \mathbf{b}_1 + \mathbf{b}_3 = \frac{4\pi}{a}\hat{\mathbf{y}}.$$

$$\boxed{\text{Reciprocal lattice: BCC, conventional cubic edge } 4\pi/a.} \quad (8)$$

(c) Phonon-mode counting along [100]

Two atoms in the basis $\Rightarrow 3 \times 2 = 6$ branches per $\mathbf{k}$: 3 acoustic, $3(2-1) = 3$ optical. 3 acoustic + 3 optical.

Along [100], C_{4v} symmetry: 1 LA + 2 TA. The two TA modes are related by the four-fold rotation about [100], hence degenerate. 1 LA, 2 (degenerate) TA $\Rightarrow$ two distinct acoustic velocities, $v_{\text{LA}} \neq v_{\text{TA}}$.

(d) LA phonon energy at $\mathbf{k} = (2\pi/a, 0, 0)$ in Si

Approximation: 1D monatomic chain along [100] with nearest-neighbour spacing $a/2$ (Si has atoms at 0 and $a/2$ projected onto [100]).

Diatomic-chain dispersion:

$$\omega(k) = 2\sqrt{K/m} \left| \sin(ka_{\text{ch}}/2) \right|, \quad a_{\text{ch}} = a/2.$$

Match small- k to sound speed:

$$\omega \approx \sqrt{K/m} k a_{\text{ch}} \equiv v_L k.$$

Read off the prefactor:

$$\omega(k) = \frac{2v_L}{a_{\text{ch}}} \left| \sin(ka_{\text{ch}}/2) \right| = \frac{4v_L}{a} \left| \sin(ka/4) \right|.$$

Evaluate at $k = 2\pi/a$:

$$\omega = \frac{4v_L}{a} \sin(\pi/2) = \frac{4v_L}{a}.$$

Energy:

$$\hbar\omega = \frac{4\hbar v_L}{a} = \frac{4 \cdot 1.0546 \times 10^{-34} \cdot 8433}{0.5431 \times 10^{-9}} \text{ J}.$$

$$\hbar\omega \approx 6.55 \times 10^{-21} \text{ J} \approx 0.041 \text{ eV}.$$

$$\boxed{\hbar\omega_{\text{LA}}(2\pi/a, 0, 0) \approx 40 \text{ meV.}} \quad (9)$$

Approximations: 1D chain replaces 3D dynamics; sinusoidal dispersion extrapolated from sound speed; harmonic. Experimental Si LA at X is $\sim 50 \text{ meV}$, agreement within a factor of ~ 1.2 .

(e) Why ALP and Si differ at $\mathbf{k} = (2\pi/a, 0, 0)$

In Si the two basis atoms are identical: nonsymmorphic diamond symmetry forces $\text{LA}(X) = \text{LO}(X)$ and $\text{TA}(X) = \text{TO}(X)$ (degenerate at the zone boundary).

In ALP the two basis atoms differ in mass ($M_{\text{Al}} = 26.98 \text{ u}$, $M_{\text{P}} = 30.97 \text{ u}$). 1D diatomic chain at the zone boundary:

$$\omega_{\text{LA}}^2 = \frac{2K}{M_{\text{max}}}, \quad \omega_{\text{LO}}^2 = \frac{2K}{M_{\text{min}}}.$$

Mass mismatch opens a gap:

$$\omega_{\text{LO}}^2 - \omega_{\text{LA}}^2 = 2K \left(\frac{1}{M_{\text{min}}} - \frac{1}{M_{\text{max}}} \right) > 0.$$

LA mode: heavy P oscillates, Al fixed. LO mode: light Al oscillates, P fixed.

Additionally ALP is partly ionic, so long-range Coulomb dipole field gives extra LO–TO splitting (absent in covalent Si). Si: $\text{LA}(X) = \text{LO}(X)$. ALP: gap opens (LO above LA) from Al/P mass mismatch + ionicity.

Question 3 — 2D heat capacity: phonons, free electrons, gapless semiconductor

Problem. (a) 2D insulator: C_v/N_a at high T and trend on cooling. (b) 2D electron gas at $k_B T \gg E_F$: C_v/N_e and trend on cooling. (c)–(d) 2D undoped semiconductor with $E_{\text{gap}} \rightarrow 0$, $m_e^* = m_h^* \equiv m^*$: density of states; $N_c(T)$, C_v^{tot}/N_c ; reconcile with equipartition. (e) Ratio C_v^{tot}/C_v when $k_B T \ll E_F$ in (b) but $k_B T \gg E_{\text{gap}}$ in (c)/(d). Useful: $\int_0^\infty dx/(e^x + 1) = \ln 2$, $\int_0^\infty x dx/(e^x + 1) = \pi^2/12$.

(a) 2D insulator

Each 2D atom: 2 kinetic + 2 potential quadratic degrees of freedom $\Rightarrow$ equipartition $4 \cdot \frac{1}{2} k_B = 2k_B$ per atom.

$$(C_v/N_a)_{T \rightarrow \infty} = 2k_B. \quad (10)$$

2D Debye: $C_v \propto T^2$ at low T . Decreases on cooling.

(b) 2D classical electron gas, $k_B T \gg E_F$

Equipartition (2 translational kinetic d.o.f.):

$$(C_v/N_e)_{k_B T \gg E_F} = k_B. \quad (11)$$

Sommerfeld at $k_B T \ll E_F$: $C_v \propto T \rightarrow 0$. Decreases on cooling.

(c) 2D density of states

Set band edges at $E = 0$ (gap $\rightarrow 0$). Parabolic 2D dispersions:

$$E_c(\mathbf{k}) = +\frac{\hbar^2 k^2}{2m^*}, \quad E_v(\mathbf{k}) = -\frac{\hbar^2 k^2}{2m^*}.$$

2D parabolic band DOS (per area, including 2 spins) is constant:

$$g_c(E) = g_v(E) = \frac{m^*}{\pi \hbar^2} \quad (\text{each band}). \quad (12)$$

(d) $N_c(T)$, C_v^{tot}/N_c , equipartition check

Particle-hole symmetry ($m_e^* = m_h^*$) pins $\mu = 0$ at all T .

Conduction-band density per area:

$$n_c = \int_0^\infty dE \frac{m^*}{\pi \hbar^2} \frac{1}{e^{E/k_B T} + 1}.$$

Substitute $x = E/k_B T$:

$$n_c = \frac{m^* k_B T}{\pi \hbar^2} \int_0^\infty \frac{dx}{e^x + 1}.$$

Use $\int = \ln 2$:

$$N_c(T) = A \frac{m^* k_B T \ln 2}{\pi \hbar^2}. \quad (13)$$

Conduction-band thermal energy per area:

$$u_c = \int_0^\infty dE g_c E \frac{1}{e^{E/k_B T} + 1}.$$

Substitute $x = E/k_B T$:

$$u_c = \frac{m^*(k_B T)^2}{\pi \hbar^2} \int_0^\infty \frac{x dx}{e^x + 1}.$$

Use $\int = \pi^2/12$:

$$u_c = \frac{m^*(k_B T)^2}{\pi \hbar^2} \cdot \frac{\pi^2}{12}.$$

By particle-hole symmetry, valence band contributes equally:

$$u_{\text{tot}} = 2u_c = \frac{\pi m^* k_B^2 T^2}{6 \hbar^2}.$$

Differentiate:

$$\frac{C_v^{\text{tot}}}{A} = \frac{\partial u_{\text{tot}}}{\partial T} = \frac{\pi m^* k_B^2 T}{3 \hbar^2}. \quad (14)$$

Divide by n_c from (13):

$$\frac{C_v^{\text{tot}}}{N_c} = \frac{\pi m^* k_B^2 T / (3 \hbar^2)}{m^* k_B T \ln 2 / (\pi \hbar^2)}.$$

Cancel:

$$\boxed{\frac{C_v^{\text{tot}}}{N_c} = \frac{\pi^2}{3 \ln 2} k_B \approx 4.74 k_B.} \quad (15)$$

No equipartition violation: at fixed T there are $\sim N_c$ electrons *and* $\sim N_c$ holes; the heat capacity is shared between $\sim 2N_c$ thermally active particles, not N_c . Moreover the carrier number itself depends on T (it grows linearly), a regime equipartition does not address.

(e) Ratio C_v^{tot}/C_v

2D Fermi gas at $k_B T \ll E_F$ (Sommerfeld with constant DOS $g(E_F) = m/\pi \hbar^2$):

$$\frac{C_v}{A} = \frac{\pi^2}{3} g(E_F) k_B^2 T = \frac{\pi m k_B^2 T}{3 \hbar^2} \cdot \pi.$$

Simplify:

$$\frac{C_v}{A} = \frac{\pi^2 m k_B^2 T}{3 \hbar^2}.$$

Take ratio (with $m^* = m$):

$$\frac{C_v^{\text{tot}}}{C_v} = \frac{\pi m^* k_B^2 T / (3 \hbar^2)}{\pi^2 m k_B^2 T / (3 \hbar^2)}.$$

Cancel:

$$\boxed{\frac{C_v^{\text{tot}}}{C_v} = \frac{1}{\pi} \approx 0.318.} \quad (16)$$

Gapless semiconductor has more heat per carrier but far fewer carriers, so per unit area the Fermi gas wins by π .

Question 4 — Magnetism of GaAs and shallow donors

Problem. (a) Magnetic class of undoped GaAs at $T = 0$. (b) Larmor–Langevin $\chi = -\mu_0 \rho e^2 \langle r^2 \rangle / (6m_e)$ for undoped GaAs. (c) Light n -doping with Si: binding radius, binding energy, and the temperature at which donor spin and orbital susceptibilities cancel. (d) Below density $1/a_{\text{bind}}^3$ the impurity-band width is $W = E_{\text{bind}} \exp(-\alpha n^{-1/3}/a_{\text{bind}})$; explain the exponential. (e) At $T = 0$, write $\chi_{\text{spin}}^{\text{IB}}(n)$ with $\alpha = 1$; find the dopant density at which it equals $|\chi_{\text{LL}}|$ from (b), and the upper temperature for validity. GaAs data: $\rho_m = 5.9 \text{ g cm}^{-3}$, $E_g = 1.4 \text{ eV}$, $m^* = 0.067m_e$, $g = -0.45$, $\epsilon_r = 12.9$.

(a) Magnetic class

Filled valence band, empty conduction band, no localised moments, no Pauli paramagnetism (zero DOS at E_F), no exchange. Only orbital response (Lenz): Diamagnetic.

(b) Larmor–Langevin χ for undoped GaAs

GaAs formula-unit mass:

$$m_{\text{f.u.}} = (69.7 + 74.9) \text{ u} = 144.6 \text{ u}.$$

Number density of formula units:

$$n_{\text{f.u.}} = \frac{5.9 \times 10^3}{144.6 \cdot 1.6605 \times 10^{-27}} \text{ m}^{-3} = 2.46 \times 10^{28} \text{ m}^{-3}.$$

Electrons per formula unit: $Z_{\text{Ga}} + Z_{\text{As}} = 64$.

Electron density:

$$\rho_e = 64 n_{\text{f.u.}} \approx 1.57 \times 10^{30} \text{ m}^{-3}.$$

Approximation (valence dominant, 30% level): $\langle r^2 \rangle \sim a_0^2$:

$$\langle r^2 \rangle \approx (0.529 \times 10^{-10})^2 \text{ m}^2 \approx 2.8 \times 10^{-21} \text{ m}^2.$$

Substitute into $\chi = -\mu_0 \rho_e e^2 \langle r^2 \rangle / (6m_e)$:

$$\chi_{\text{LL}} = -(4\pi \times 10^{-7}) \cdot \frac{1.57 \times 10^{30} \cdot (1.602 \times 10^{-19})^2}{6 \cdot 9.109 \times 10^{-31}} \cdot 2.8 \times 10^{-21}.$$

Evaluate the inner factor:

$$\frac{1.57 \times 10^{30} \cdot 2.566 \times 10^{-38}}{5.466 \times 10^{-30}} \approx 7.37 \times 10^{21}.$$

Multiply by μ_0 and $\langle r^2 \rangle$:

$$\chi_{\text{LL}} \approx -(1.257 \times 10^{-6}) \cdot 7.37 \times 10^{21} \cdot 2.8 \times 10^{-21}.$$

Numerical:

$$\chi_{\text{LL}} \sim -3 \times 10^{-5} \text{ (SI, dimensionless).} \quad (17)$$

Order-of-magnitude consistent with tabulated values for covalent semiconductors.

(c) Donor binding radius, energy, and cancellation T^*

Hydrogenic with $e^2 \rightarrow e^2/\epsilon_r$, $m_e \rightarrow m^*$:

$$a_{\text{bind}} = \frac{4\pi\epsilon_0\epsilon_r\hbar^2}{m^*e^2} = a_0 \frac{\epsilon_r m_e}{m^*}.$$

Evaluate:

$$a_{\text{bind}} = 0.529 \text{ \AA} \cdot \frac{12.9}{0.067} = 192.5 a_0 \approx 10.2 \text{ nm}.$$

$$E_{\text{bind}} = \text{Ry} \cdot \frac{m^*/m_e}{\epsilon_r^2} = 13.6 \text{ eV} \cdot \frac{0.067}{12.9^2}.$$

Evaluate:

$$E_{\text{bind}} \approx 13.6 \text{ eV} \cdot 4.03 \times 10^{-4} \approx 5.5 \text{ meV}.$$

$$\boxed{a_{\text{bind}} \approx 10.2 \text{ nm}, \quad E_{\text{bind}} \approx 5.5 \text{ meV}.} \quad (18)$$

Donor spin (Curie, $S = \frac{1}{2}$):

$$\chi_{\text{spin}} = \frac{\mu_0 n_d g^2 \mu_B^2}{4k_B T}.$$

Donor orbital (Larmor with effective mass m^* , hydrogenic $\langle r^2 \rangle_{1s} = 3a_{\text{bind}}^2$):

$$\chi_{\text{orb}} = -\mu_0 \frac{n_d e^2}{6m^*} \cdot 3a_{\text{bind}}^2 = -\frac{\mu_0 n_d e^2 a_{\text{bind}}^2}{2m^*}.$$

Set $\chi_{\text{spin}} + \chi_{\text{orb}} = 0$:

$$\frac{\mu_0 n_d g^2 \mu_B^2}{4k_B T^*} = \frac{\mu_0 n_d e^2 a_{\text{bind}}^2}{2m^*}.$$

Cancel $\mu_0 n_d$:

$$k_B T^* = \frac{g^2 \mu_B^2 m^*}{2e^2 a_{\text{bind}}^2}.$$

Use $\mu_B^2 = e^2 \hbar^2 / (4m_e^2)$:

$$k_B T^* = \frac{g^2 \hbar^2 m^*}{8m_e^2 a_{\text{bind}}^2}.$$

Use $E_{\text{bind}} = \hbar^2 / (2m^* a_{\text{bind}}^2)$, so $\hbar^2 / (m^* a_{\text{bind}}^2) = 2E_{\text{bind}}$:

$$k_B T^* = \frac{g^2}{8} \cdot \left(\frac{m^*}{m_e} \right)^2 \cdot 2E_{\text{bind}}.$$

Simplify:

$$\boxed{k_B T^* = \frac{g^2}{4} \left(\frac{m^*}{m_e} \right)^2 E_{\text{bind}}.} \quad (19)$$

Numerical with $g^2 = 0.2025$, $(m^*/m_e)^2 = 4.49 \times 10^{-3}$, $E_{\text{bind}} = 5.5 \text{ meV}$:

$$k_B T^* \approx \frac{0.2025 \cdot 4.49 \times 10^{-3} \cdot 5.5 \times 10^{-3}}{4} \text{ eV}.$$

Evaluate:

$$k_B T^* \approx 1.25 \times 10^{-6} \text{ eV}.$$

$$\boxed{T^* \approx 15 \text{ mK}.} \quad (20)$$

(d) Why the impurity bandwidth is exponentially small

Donor 1s wavefunction decays as $e^{-r/a_{\text{bind}}}$. Tight-binding hopping t between donors at separation $r \sim n^{-1/3}$ is the wavefunction overlap, also $\propto e^{-r/a_{\text{bind}}}$. Bandwidth $W \sim t$, with prefactor E_{bind} (the only available energy scale):

$$W \sim E_{\text{bind}} \exp\left(-\alpha n^{-1/3}/a_{\text{bind}}\right).$$

Exponential = wavefunction overlap of neighbouring 1s donor states.

(e) Impurity-band spin χ , crossover density, validity

For $k_B T \ll W$ the half-filled impurity band is a Pauli system. DOS $\sim n/W$ (one state per donor in band of width W):

$$\chi_{\text{spin}}^{\text{IB}} \sim \mu_0 (g\mu_B)^2 \frac{n}{W}.$$

Substitute W with $\alpha = 1$:

$$\chi_{\text{spin}}^{\text{IB}} \sim \mu_0 \frac{g^2 \mu_B^2 n}{E_{\text{bind}}} \exp\left(\frac{1}{n^{1/3} a_{\text{bind}}}\right). \quad (21)$$

Crossover $\chi_{\text{spin}}^{\text{IB}} \sim |\chi_{\text{LL}}| = 3 \times 10^{-5}$. Evaluate the prefactor at $n^* \sim 1/a_{\text{bind}}^3 \approx 10^{24} \text{ m}^{-3}$:

$$\mu_0 (g\mu_B)^2 = (1.257 \times 10^{-6}) \cdot 0.2025 \cdot (9.27 \times 10^{-24})^2 \approx 2.19 \times 10^{-53}.$$

Divide by $E_{\text{bind}} = 8.8 \times 10^{-22} \text{ J}$:

$$\frac{\mu_0 g^2 \mu_B^2}{E_{\text{bind}}} \approx 2.5 \times 10^{-32} \text{ m}^3.$$

Multiply by $n = 10^{24} \text{ m}^{-3}$:

$$\frac{\mu_0 g^2 \mu_B^2 n}{E_{\text{bind}}} \approx 2.5 \times 10^{-8} \quad \text{at } n = 1/a_{\text{bind}}^3.$$

Match $|\chi_{\text{LL}}| = 3 \times 10^{-5}$ requires exponential $\sim 10^3$:

$$n^{-1/3}/a_{\text{bind}} \sim \ln(10^3) \approx 7.$$

Solve:

$$n^* \sim \frac{1}{(7a_{\text{bind}})^3} \sim 3 \times 10^{-3}/a_{\text{bind}}^3 \sim 3 \times 10^{21} \text{ m}^{-3} \sim 3 \times 10^{15} \text{ cm}^{-3}.$$

$$n^* \sim 10^{15}\text{--}10^{16} \text{ cm}^{-3} \quad (\text{well below the Mott density } \sim 10^{18} \text{ cm}^{-3}). \quad (22)$$

Validity: Pauli regime requires $k_B T \ll W$. At n^* :

$$W \sim E_{\text{bind}} e^{-7} \approx 5.5 \text{ meV} \cdot 9 \times 10^{-4} \approx 5 \text{ } \mu\text{eV}.$$

$$T_{\text{max}} \sim W/k_B \sim 50 \text{ mK--}0.1 \text{ K}. \quad (23)$$

Sub-kelvin temperatures (dilution-fridge regime).

End of solutions.